

Lean quality improvement model for quality practices in software industry in Pakistan

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ABSTRACT

Implementation of quality and achieving quality culture in small and medium software houses (SMSh) have been a subject of discussion among the industry. The existing software process improvement frameworks are too heavy for SMSh. There is a need of lean quality models that will help SMSh in establishing quality culture with minimal effort and resources. The objective of this research study is to map the environment and culture of SMSh in Pakistan towards quality improvement and process improvement by implementing total quality management philosophy. A lean quality improvement model (LQIM) consisting of four quality constructs and 10 quality practices has been proposed. The LQIM is validated using good fit indices in structural equation modeling. At the end, implementation of the proposed LQIM is explained using the Deming's philosophy of plan, do, check, act cycle for continuous process improvement. Copyright © 2015 John Wiley & Sons, Ltd.

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KEY WORDS: software process improvement; quality culture; TQM; quality models; quality; practices; lean quality

1. INTRODUCTION

Quality is not just the responsibility of one person in an organization; it is the duty of all the employees. Everyone involved directly or indirectly in the development of a software product is responsible for its performance and services. Failure of the management to implement satisfactory quality methodology and non-interest of the employees to execute processes rigorously are among the reasons for failure of quality implementation and collapse of the quality practices in an organization [1]. The management and the operational staff are reluctant to include quality assurance activities in their production processes as they presume quality as an overhead [1, 2]. There is a very powerful statement by the quality guru Philip Crosby, 'Quality is free', meaning there is a cost for poor quality, which is the expense of doing things wrong [3]. Thus, what is needed is a system that ensures that all procedures that have been designed and planned are followed.

As asserted by Huarng *et al.* [4], maximum benefit of quality standards implementation in organizations can be achieved through adopting the philosophy and practices of total quality management (TQM). 'In today's dynamic era of globalization, customer retention is only achieved through TQM and continuous improvement' [5]. TQM program provides a paradigm shift in management philosophy for improving organizational effectiveness [6]. By implementing TQM

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principles, not only it motivates employees but also gives new direction and quality guidelines to an organization to achieve customer satisfaction through continuous improvement [7, 8].

Software practitioners in developing countries have realized the importance of quality in software development, but still implementation of TQM practices is lacking in the software industry. Software managers give more emphasis to cost reduction and functionality rather than practicing quality process improvement [9]. In order to improve quality, a more proactive approach is required where preventive measures may be introduced by following TQM philosophy that will not only minimize the frequency of defect occurrence but will also remove the causes of defects from source.

The quality models that are developed in context of large organizations tend to 'introduce a degree of mechanization and inflexibility that small and medium enterprises (SME) find difficult to accept in times of turbulent market changes, which demand increased agility' [10]. Leading models like capability maturity model (CMM) [11] and CMM integration (CMMI) [12] are designed to suit requirements of large organizations, and the time and cost of implementation and training are very high. It is possible to have transition from CMM to International Organization for Standardization (ISO) 9000; however, it is argued that ISO 9000 may only be helpful to induce quality awareness and is not suitable enough for software process improvement (SPI) [13]. ISO international standards are hard to apply in small projects, in small organizations, and in companies that have employees in between 1 and 25 [14]. While reviewing SPI models for SMEs, Mishra and Mishra [15] did not consider CMM [11], ISO/International Electrotechnical Commission (IEC) 9003:2004 [16, 17], the TickIT [18], Bootstrap [19], and Initiating, Diagnosing, Establishing, Acting and Learning (IDEAL) [20] appropriate. In their view, these models were not appropriate for SMEs having small size and limited resources.

This study takes a research-based perspective to propose a lean model for quality improvement, in light of proactive defect prevention approach following TQM philosophy and Deming's plan, do, check, act (PDCA) cycle [21]. The focus is to minimize the frequency of defect occurrence by removing the causes of defects at source by introducing preventive measures throughout the software development life cycle (SDLC) in light of the practices and perceptions of the local Information Technology (IT) practitioners of small and medium software houses (SMSh). Thus, the question addressed in this study is 'What can be a proposed SPI model, which can best fit to solve the problems of quality improvement in the local small and medium software houses?'

Firstly, a theoretical framework was developed for the study that was based on two quality management models according to the best practices available in the industry. Then a data collection instrument was developed and pretested for the study for its reliability. It was then administered for data collection from SMSh so that their feedback and experience of the quality improvement practices could be incorporated in the proposed lean quality improvement model (LQIM). The effectiveness of the proposed lean model has been statistically validated using both the confirmatory and exploratory modeling validating the design and development of theory using structured equation modeling (SEM) technique [22]. Application of statistical techniques to such problems is common in the field of management science [23]. SEM is an effective and universally acceptable tool to benchmark and validate model effectiveness [24–28]. Therefore, it has been used to validate LQIM effectiveness according to the practices of SMSh. The statistical analysis shows good model fit results based on model fit indices: goodness-of-fit index (GFI), comparative fit index (CFI), and root mean square error (RMSEA).

This paper is divided into the following sections. Section 2 looks at related work and presents different types of leading quality models that are practiced in the software industry. Section 3 presents the research methodology adopted. In Section 4, empirical analysis is presented to develop and validate using SEM an LQIM for SMSh in Pakistan. In Section 5, the proposed LQIM is presented based on Deming's TQM philosophy of PDCA [21, 29] along with a deployment plan for implementation of TQM practices. Section 7 concludes the paper.

2. RELATED WORK

In order to achieve excellence in software product development and to produce reliable, efficient, maintainable, and highly optimized software products, IT practitioners are becoming well aware of

quality and SPI. The motivation behind the quality oriented preventive practices is to reduce rework cost through minimization of errors. It has resulted into introduction of new wave of products for process modeling, evaluation, and improvement [30]. In the following sections, different leading quality models that are practiced in the software industry are presented.

2.1. *Quality models*

In the late 1940s, the father of the Toyota Production System, Taiichi Ohno, tossed the term *lean manufacturing principle: eliminating waste* [31]. Lean product development (LPD) is a model that is adopted from Lean Manufacturing and Lean IT Principles and Practices to the project development domain. It advocates to eliminate waste and to induce learning culture, team empowerment, holistic approach, and system integrity. Another technique of lean product development is value stream mapping, which is used to identify and strip wasteful steps out of main process flow of information and resource allocation [32]. The end goal of this technique is to make the company 'lean', meaning free of wasted effort [33].

In one of his lectures in 1950 at the Union of Japanese Scientists and Engineers seminar in Japan, Dr. Edward Deming for first time coined the term plan, do, study, act cycle as modification of Dr. Walter Shewhart's [34] PDCA cycle [21]. Dr. Deming stressed on the notion that four steps of design, production, sales, and redesign should be constantly rotated through market research with the aim of improvement in quality of product and service. Deming's PDCA cycle evolved into an overall quality improvement model, which emphasized on building new knowledge to have better hold on changes and improvements [35, 36].

Since 1990s, the emphasis on managing the projects, quality standards, and software process models has increased. ISO [37] established ISO 9001 generalized standards for quality management systems suitable for industrial processes. Later in 1995, ISO and IEC jointly released ISO/IEC 12207, a standard for information technology life cycle processes. For software process measurement, ISO/IEC 15939 was developed. ISO/IEC 90003:2004 is the guideline for the application of ISO 9001:2000 to computer software [17]. British Standard Institute initiated its own standard guidelines to implement ISO 9001. Companies who satisfy the British Standard Institute guidelines obtain ISO 9000/TickIT certification [38].

The Software Engineering Institute (SEI) of Carnegie Mellon University developed a framework called CMM. CMM v.1.0 for software was released in 1991, which provided a comprehensive framework with detailed description of processes for the software development and system engineering [39]. Later, SEI rereleased CMM version 1.1 in 1992, which was tailored according to the increasing needs of the software industry [11, 40]. The latest version is CMMI version 1.3 [12].

Software Engineering Institute also developed team software process (TSP) [41] and personal software process (PSP) [42] to facilitate superior performance of software development teams and individuals in SMEs, respectively. Both the TSP and PSP help ensure quality software products and improve process management in SMEs.

Software quality improvement models contribute towards greater customer satisfaction, focus on improving process to gain maximum value for money, and increase return on investment in shortest possible time. A summary roadmap of the evolution of quality models in the fields of engineering, software, IT, and manufacturing has been given by Sheard [38].

2.2. *Quality models for small organizations*

Existing software quality improvement models are not easily applicable for SMEs because of lack of dedicated quality professionals, lack of quality domain knowledge, and lack of basic infrastructure to implement quality in practice [43]. Attempts have been made to understand the dynamics of the SMEs and to propose lean models for SPI so that product quality can be improved. According to Damm *et al.* [44], fault detection in the early phases of SDLC is more cost effective. Practice of ignoring faults not only increases rework cost but also increases the risk of project delays. Robinson *et al.* [45] talk about improving quality through planning defect improvement process through defect removal prioritization strategy. Defect classification by SDLC effectively reduces rework cost, improves quality, and gives better results as compared with sequential last minute testing. Russ and

McGregor [46] describe a software development process for small projects. The proposed process integrates the basic iterative, incremental model with a guided inspection technique [47], and the measurement framework provided by Humphrey [48].

Walloon Observatory for Software Practices (OWPL) [49] is a three-stage SPI approach that allows SMEs to start small within a limited budget and reach an acceptable level according to SPI standard models. Software process matrix model for small organizations [50, 51] is based on quality function deployment and helps an organization in finding relative importance of software processes by prioritizing them using a matrix. Allen *et al.* [52] describe a process model called process improvement for small to medium software enterprises (PRISMS) suitable for SPI in small to medium organizations. Key features of the PRISMS process are ‘explicitly relating improvement goals to business goals, which are used to choose and prioritize key process areas for improvement; the use of a simplified and tailorable assessment method based on CMM; the definition and use of metrics from the outset; and mechanisms to encourage maximum transparency in the process and participation by all stakeholders’ [52]. Pettersson *et al.* [53] present improvement framework utilizing lightweight assessment and improvement planning, a step-by-step guide to lightweight process assessment and improvement aimed at practitioners undertaking SPI initiatives. It involves prioritizing improvement activities, identifying dependencies, and thus establishing a realistic quality improvement plan. Petersen and Wohlin [54] discuss SPI through a lean measurement method called SPI-LEAM. Wang *et al.* [55] provide an experience report of the application of lean approaches in agile software development.

Main focus of these models is to follow a corrective approach of defect management through prioritization, including defect removal prioritization strategy and prioritization of key process areas for improvement based on performance matrices. A comparative view of a selected set of SPI models for SMEs has been given by Mishra and Mishra [15]. They have presented and discussed significant characteristics of these models and highlighted their constraints too. The major constraint is the need to have experienced quality assessment and measurement experts to analyze and prioritize process improvement activities.

3. METHODOLOGY

This research was carried out through a survey using a questionnaire as a data collection instrument to obtain the information about the process improvement and quality management practices being followed by the local SMSH. The questionnaire was pretested with a small sample of 30 respondents. It provided the initial information that helped to narrow down the scope of the research. It also gave an insight into the local industry’s quality improvement practices. For the purpose of this research, an organization having 50 or less employees is considered as a small organization, and an organization having employees between 50 and 250 is considered a medium organization [56, 57]. The following sections describe the methodology adopted for this research study.

3.1. *Quality constructs and theoretical framework*

Researchers have identified factors that are generally considered critical for successfully implementing SPI programs [58–60]. Different researchers have attempted to investigate the quality constructs in TQM with various rationales and objectives. Main rationale of Saraph *et al.* [61] was to develop a data collection tool to measure quality management practices of companies. Black and Porter [62] developed their quality constructs from the Malcolm Baldrige National Quality Award (MBNQA) criteria [63], on the pretext that MBNQA is a well recognized and established quality framework. Tamimi and Gershon [64] also developed a data collection tool by following Deming’s 14 points. Ahire *et al.* [65] proposed a validation instrument based on 12 quality implementation constructs, derived from literature, to measure quality management practices for manufacturing industry. Similarly, Dutta *et al.* [66] designed a data collection instrument for a European study based on quality assessment models like CMM [67], Software Process Improvement and Capability Determination (SPICE) [68], and Bootstrap [19]. Dyba [69] presented key factors of success in SPI along with an instrument for measuring them. According to

these studies, the authors tried to systematically present quality constructs to measure quality management and to address the problems faced during TQM adoption process.

In this study, through literature review and analysis, a set of eight quality constructs were developed to measure quality management. These quality constructs are mapped onto performance indicators for quality management as shown in Table I. Table II gives a summary of quality constructs reported in literature by different authors. Using the developed quality constructs, a framework was proposed to find out a minimal set of most significant best practices for the implementation of quality management paradigm in the local industry. In this framework, the process quality improvement (PQI) construct is the dependent variable, and all other constructs are independent variables.

For this study, the performance indicators for quality management were derived from the clauses of ISO 9000 [76, 77, 16, 17] and CMM key process areas [11] as shown in Table I. These indicators were then used to formulate questionnaire statements. A total number of 48 question statements were prepared with a purpose to depict the level of understanding and implementation of quality standards among SMSH. A 5-point Likert scale was used for the data collection. The questionnaire design technique followed is similar to that of Black and Porter [62] who used MBNQA to develop TQM constructs. MBNQA along with the Singapore Quality Award criteria has also been used by Quazi and Padibjo [78] to design questionnaire and to study the impact of training and development on ISO 9000 certified companies in Singapore.

3.2. Validity

This section discusses threats to validity. Validity means that if the instrument measures what was intended to measure, then that instrument is simply valid [79, 80]. According to Wohlin *et al.* [81], validity can be discussed in terms of construct, internal, external, and conclusion validity. Construct validity that is concerned with mapping of the real world to laboratory is not relevant here as the research study described in this paper was carried out based on data collected from practitioners from industry [53].

3.2.1. Internal validity. According to Runeson and Host [80], internal validity denotes the trustworthiness of results and extent to which these results are true and not biased. Internal validity measures the quality of answers provided against the research questions. Internal validity of the data and data collection instrument includes the determination of cause and effect relationships introduced by Amaratunga and Baldry [82]. In order to ensure internal validity, respondents were assured of the anonymity and confidentiality of data so that they could express their opinion without any hesitations.

3.2.2. External validity. External validity investigates whether the research findings in a particular environment can be generalized for other situations in which a sample population is investigated [80]. Such a generalized behavior can be a critical characteristic of research that further signifies the research scope and makes it contributory to multidimensional fields and body of knowledge [82]. To further enhance the external validity of the research, a systematic random sampling technique was adopted to select an unbiased sample size from a list containing software houses located mainly in four major cities of Pakistan. The data were collected from members of the Pakistan Software Export Board [83] and Pakistan Software Houses Association [84] and other software houses. Generalizing the results across different cultures and environments was not the aim of this study. It is envisaged from the earlier discussion that this research has fulfilled the prerequisites and has good external validity.

3.2.3. Conclusion validity and reliability. The sampling technique used can pose a threat to the conclusions drawn. This threat was controlled by developing a representative list of software houses through multiple sources and professional bodies representing the SMSH as emphasized by Stake [85]. Then systematic random sampling technique was used to select an unbiased sample size from this list. Data quality was checked statistically to filter the questionnaires that were skewed at extremes. Partially filled questionnaires were left out of the analysis. Questionnaire responses having all '1' or all '2', or all '3' or all '4' or all '5' were treated as constants and were not included in the analysis.

Reliability of data is determined from practical considerations. In order to further clarify the concepts and interpretations, all questions were pretested by discussing them with several IT professionals, in an attempt to eliminate biases of ambiguity [86] and replication of questions.

Table I. Constructs and performance indicators.

Constructs	Performance indicator	Variable	Reference clause (ASQ 2000-ISO 9000/CMM)
Organization size and structure	Size	SIZ	0.1/CMM-KPA-L2
	Technology, research, and development	TND	6.3/CMM-KPA-L5
	Statistician	STA	CMM
	Organization Structure	OST	0.1
Organization culture	Retention	RTN	5.1/CMM-KPA-L5
	Overtime	OVT	6
	Scheduling	SHD	6/CMM-KPA-L2
	Learning	LRN	6.2.2
	Communication	COM	5.5/7.5.5/CMM-KPA-L3
	Time management	TM	6/CMM-KPA-L5
	Training	TRA	6.2.27.5.5/CMM-KPA-L3
	Team structure	TST	4/CMM-KPA-L2
	Turnover rate	TR	5.1/CMM-KPA-L5
	Assessment	ASS	6.2.2/4.2.4
Organizational behavior towards quality	Resource allocation	RA	6.1/CMM-KPA-L2
	Turnover rate	TR	5.1/CMM-KPA-L5
	Software quality improvement	SQI	7.3.7/CMM-KPA-L5
	Quality assurance	QA	7.3.4/CMM-KPA-L2
Requirement development and management	Top management quality approach	TPQ	6.1/CMM-KPA-L3
	Project review	PR	7.1/CMM-KPA-L2
	Estimation	EST	7.3.4/CMM-KPA-L2
	Customer satisfaction	CS	8.2.1/CMM-KPA-L4
Project planning	Change management	CM	7.3.7/CMM-KPA-L5
	Outsourcing	OSR	CMM-KPA-L2
	Conformance	CNF	7.3.6/7.5.5/CMM-KPA-L2
	Acceptance testing	UAT	7.3.5/7.3.6/CMM-KPA-L3
	Project planning	PPL	7.3.1/CMM-KPA-L2
	Risk management	RM	8.5.2/8.5.3/CMM-KPA-L2
	Coding standards	CSTD	8.2.2a
	Audits and review	AR	5.6/8.2.2
Project monitoring and control	Project monitoring	PM	8.2.3/8.2.4/CMM-KPA-L2
	Project tracking	PT	8.2.3/8.2.4/CMM-KPA-L2
	Performance	PER	8.2.3
	Team work	TW	4/CMM-KPA-L2
Measurement and analysis	Process measurement	PMR	8.2.3/CMM-KPA-L3
	Data collection system	DCS	8.4/CMM-KPA-L4
	Process performance	PPR	8.2.3/CMM-KPA-L4
	Defect log	DL	8.4/CMM-KPA-L5
	Knowledge base library	KBL	8.4/CMM-KPA-L2, L3
Process quality improvement	Resource planning	RP	6.2/CMM-KPA-L2
	Corrective action	CA	8.5.1
	Total quality management	TQM	7.3
	Quality planning	QPL	5.4.2a/CMM-KPA-L4
	Continuous process improvement	CPI	8.5.1/CMM-KPA-L3
	Organization commitment	OCM	5.1/CMM
	Configuration management	SCM	CMM-KPA-L2
	Process improvement	SPI	6.2.1/CMM-KPA-L3
	Maturity level	ML	CMM

L, CMM level; KPA, key process area; CMM, capability maturity model; ASQ, American Society for Quality; ISO, International Organization for Standardization.

Purpose and objectives were clearly stated in the cover letter from the researcher. In order to ensure confidentiality of the respondents as advised by Patton [87], respondents were reminded not to fill names or email addresses. Assurances were also given to respondents about confidentiality and

Table II. Quality constructs reported in literature.

Quality constructs	Saraph <i>et al.</i> [60]	Flynn <i>et al.</i> [70]	Ahire <i>et al.</i> [65]	Raoa <i>et al.</i> [71]	Yusof and Aspinwall [72]	Parast <i>et al.</i> [73]	Sila [74]	Zakuan <i>et al.</i> [75]
Organization planning						Strategic planning		Strategic planning
Organization behavior towards quality	The role of top management leadership	Top management support	Top management commitment	Top management commitment	Management leadership	Quality leadership	Leadership	Quality leadership
Requirement management	The role of quality department	Customer involvement	Customer focus	Quality citizenship	—	Customer focus and satisfaction	Customer focus	Customer focus and satisfaction
Project monitoring and control	Product/service design	Product design	Design quality management	Customer orientation	—	—	—	Quality assurance
Measurement and analysis	Quality data and reporting	Quality information	—	Quality information availability	Measurement and feedback	Quality information and analysis	Information and analysis	Quality information and analysis
	—	—	—	Internal quality results	—	—	Financial and market results	Quality results
	—	—	—	External quality results	—	—	—	—
Process quality improvement	Process management	Process management	Statistical process control usage	Process design	System and process improvement tools	Process of quality management	Process management	—
	—	—	—	—	and techniques	Quality results	—	—
	—	—	—	—	Continuous improvement process	Quality assurance of products and services	—	—
Training	Training	Workforce management	Employee training	Employee training	Education and training	Support for human resource development	Human resource management	Human resource development
	—	—	—	—	Human resource development	—	—	—
Supplier management	Supplier quality management	Supplier involvement	Supplier quality management	Supplier quality	Supplier quality assurance	Supplier quality	Supplier management	Supplier quality management
	—	—	Supplier performance	—	—	—	—	—

anonymity [88]. To ensure reliability of data that were gathered through interviews, prior appointment was sought so that the interview sessions are conducted without interruption and without discussion with others practitioners [53]. During field visits, decorum and manners were ensured to develop the interest of the participants so that they are not bored [89, 85, 90].

3.3. Survey administration

In the first step, a list of 1031 software houses was compiled from the members of Pakistan Software Export Board [83] and Pakistan Software House Association [84]. It was ensured that there were no duplicate names or emails in the list, and it was not an ordered list. Systematic random sampling technique was adopted to collect the sample. It also ensured good geographical distribution according to population density. First, systematic sample was drawn by selecting every third name from the list. As a result, a sample of 343 names was compiled in the first run. Following the same technique, 229 and 153 names were selected in the second and third rounds of systematic selection by changing the selection interval. In this way, a total of 725 questionnaires were administered on local IT practitioners through mail, one questionnaire by post, and one by email per company for filling online survey. One hundred forty four respondents filled out complete questionnaires. The response rate through mail was 19.8%.

To further improve the response rate, data collection was also administered in person. As an additional effort during the period of 6 months, more than 90 software houses from the compiled sample list were approached from time to time. A telephonic appointment was taken before approaching the software houses. Each organization was given two to three questionnaires. The response rate was fairly good, and 83 completed questionnaires were received in this round. The total number of questionnaires completed for the study was 227 having an overall response rate of 31.3%. Four questionnaire responses having all '1' or all '2', or all '3' or all '4' or all '5' were treated as constants and were not included in the completed responses. The preceding method of survey administration was adopted to ensure the spread of responding practitioners across different organizations.

Out of the total respondents, 7% was in the top management including chair person, 39% was in the middle management including program managers and line managers, and 54% was in the lower management including software engineers, system analysts, and quality assurance personnel. It is interesting to note that 12% of the organizations to which the respondents belonged to used CMM, 7% followed CMMI, 42% followed ISO, 4% used TSP/PSP, and 35% was practicing other models. Out of these companies, 26.4% had documented subcontract management procedures for outsourcing projects to external clients. Of these companies, 81% had employees less than 200 indicating that large number of software companies of the respondents were in the category of SMEs.

4. EMPIRICAL ANALYSIS

The empirical analysis includes correlation and regression analysis of quality constructs. SEM technique was used to develop an optimized LQIM for standard quality practices in SMSH. Eight quality constructs were developed to ascertain the level of current quality practices and to evolve an LQIM. In correlation analysis, all seven independent constructs were found significant towards the dependent variable quality improvement. Regression analysis revealed that only four of these independent quality constructs contributed significantly towards the dependent variable PQI. Through SEM, the evolved LQIM presented four quality constructs and 10 of their respective quality practices as significant. In the following sections, analysis of responses against each item in the questionnaire based on eight quality constructs is presented. The Likert scale values corresponding to 1 and 2 are accumulated to represent low agreement, and Likert scale values corresponding to 4 and 5 are accumulated to represent high agreement.

In order to ascertain the reliability of data collection instrument and the data collected against each indicator variable, Cronbach's alpha [91] test was used. According to Murphy and Balzer [92], generally Cronbach's alpha value of greater than 0.70 is considered adequate. A value of 0.839 for

the Cronbach's alpha was obtained for this research based on 48 items of the questionnaire and 227 respondents. The Cronbach's test was also applied to all eight constructs individually. The Cronbach's alpha for organization size and structure (OSS) was 0.77; for organization culture (OCL) was 0.53; for organization behavior towards quality was 0.67; for requirement management, it was 0.61; for project planning, it was 0.62; for project monitoring and control (PMC), it was 0.51; for measurement and analysis (MAN), it was 0.72; and for PQI, it was 0.68. The reliability of instrument is valid and good as according to Nunnally [93] Cronbach's alpha value of more than 0.5 is also acceptable.

4.1. Correlation analysis

The correlation matrix for eight quality constructs for quality improvement and quality implementation is given in Table III where results are reported along with Pearson correlation coefficient value denoted by R . The level of significance is P and sample size is $N=227$. Correlation is considered significant at level 0.01 if P -value is near to 0.01; similarly, correlation is considered significant at level 0.05 in case P -value is near to 0.05. If the correlation is high between two variables, then it is said that the two variables have strong interrelationship characteristics. Table III shows that there is a strong correlation between the dependent variable and all other variables. There is significant correlation of the dependent variable PQI with seven constructs at significance level 0.05. The significance value of PPL is 0.486, that of requirement development and management is 0.547, that of OSS is 0.427, that of OCL is 0.502, that of PMC is 0.595 and that of MAN is 0.648, and lastly, that of organizational behavior towards quality is -0.341 , which shows inverse relationship. The subsequent step in the empirical analysis is to find out the impact in variability of dependent variable outcomes because of the independent variables.

4.2. Regression analysis

The output results of regression analysis prove that four independent variables are strong predictors of dependent variable PQI. As per results, R^2 is 0.553, which means that these constructs explain 55.3% variability of dependent variable PQI in the model. Significant predictors of PQI are OSS with a value of 0.000, OCL with a value of 0.019, PMC with a value of 0.000, and MAN with a value of 0.000. All of these are significant at P -value less than 0.05. The value also endorses that the relationship between PQIs is linear with the four predicting independent variables. The P -value of 0.000 for analysis of variance and large value of ratio of two mean squares $F=38$ signify that the model is statically significant at $\alpha=0.05$.

It can be deduced from this analysis that in order to make an effective quality improvement model and guidelines for the local industry, the following four critical success factors: OSS, OCL, PMC, and MAN, will play very important roles.

Table III. Pearson correlation coefficient R between all quality constructs.

	PPL	RDM	OSS	OCL	OBQ	PMC	MAN	PQI
PPL	1							
RDM	0.534**	1						
OSS	0.204**	0.356**	1					
OCL	0.384**	0.517**	0.283**	1				
OBQ	-0.236^{**}	-0.372^{**}	-0.488^{**}	-0.460^{**}	1			
PMC	0.513**	0.548**	0.254**	0.449**	-0.222^{**}	1		
MAN	0.514**	0.623**	0.324**	0.499**	-0.336^{**}	0.618**	1	
PQI	0.486**	0.547**	0.427**	0.502**	-0.341^{**}	0.595**	0.648**	1

PPL, project planning; RDM, requirement, development, and management; OSS, organization size and structure; OCL, organization culture; OBQ, organization behavior towards quality; PMC, project monitoring and control; MAN, measurement and analysis; PQI, process quality improvement.

**Correlation is significant at the 0.05 level (two-tailed)

4.3. Structural equation modeling

To further test the model fitness, SEM [22] technique was applied. SEM describes relationship between variables, is similar to regression modeling and factor analysis, and is effective in a way for removing multicollinearity in the model. A specialized statistical software called Analytical Movement of Structures (AMOS) [94] is used to further refine the model to come up with a lean SPI paradigm. AMOS is effective in removing multicollinearity between constructs that is the effect of one indicator or construct on the other ones [95]. AMOS has a graphical interface and is an excellent tool to use for SEM model fitting [95].

A second-order critical factor analysis was carried out using SEM to further validate the results obtained through the linear regression modeling in previous section. It tests the stability of the relationships between the measurement variables and the constructs. At this stage, all eight latent constructs are retained, and the total number of measurement variable indicators remains 48. The SEM completed four runs to reach acceptable GFIs benchmark level. During the run, four constructs got completely deleted. In total, 38 indicators were deleted from scale development process. These deleted indicators were found to be inadequate to load the model because of poor level of explained variance. The output of the model and the recommended standard quality practices are shown in Table V.

This study followed various criteria to determine the 'GF' of the measurement model according to Anderson and Gerbing [22], and Bentler and Bonett [96]. The results are shown in Table IV and are presented in the following sections.

4.4. Ratio of chi-square to degrees of freedom

Value of chi-square is usually very sensitive to sample size especially if the sample size is greater than 200. There is another chi-square statistic, which measures the ratio of chi-square to the degrees of freedom, which is used to measure the model fit. According to Kline [97] and Anderson and Gerbing [22], if the value of this ratio is less than 3, then it is a good indicator of model fit. As shown in Table IV, chi-square to d.f ratio is 2.323. This value is under the benchmark fit model; therefore, the GF of the proposed SPI model is within acceptable range.

4.5. Goodness-of-fit index

Goodness-of-fit index is an alternative to the chi-square test. Its criterion is to assess the ratio of variance that is accounted for by the approximation of the SPI model covariance. Through such estimation, it can be predicted how close the proposed model is able to replicate the covariance of population matrix [98]. It has also been found that as the value of GFI increases, the number of parameters in the model also increases. In reference to Table IV, the value of GFI is 0.926, which is greater than 0.9 benchmark; therefore, it can be confirmed that SPI model has a good model fit.

4.6. Comparative fit index

Comparative fit index by Bentler [99] calculates the degree of interrelationship between the suggested model and null model with the assumption that the latent measurement variables (indicators) have no

Table IV. Baseline comparisons model good fit indices.

Model	Index	SPI value	Acceptable value
Default model	Chi-square	88.286	
Default model	d.f	38	
Default model	Chi-square/d.f	2.323	<3
Default model	GFI	0.926	>0.9
Default model	CFI	0.85	range 0.8–0.9
Default model	RMSEA	0.077	<0.08

SPI, software process improvement; d.f, degrees of freedom; GFI, goodness-of-fit index; CFI, comparative fit index; RMSEA, root mean square error index.

significant interrelationship. This index works well even when sample size is small. The SPI model resultant value of CFI in Table IV is 0.85. According to Bollen [100], a CFI value of 0.85 represents progress and thus should be acceptable.

4.7. Root mean square error index

The RMSEA index by Steiger [101] has evolved as a good measure for a model's fit with regard to model's economy or stinginess. It chooses the parameters that would fit the population covariance matrix [102]. In other words, it assesses the divergence among the proposed and estimated covariance matrices per degree of freedom. In recent years, it has gained importance among researchers because of its sensitivity to the number of measurement parameters. According to MacCallum *et al.* [103] and McQuitty [104], if the value of RMSEA lies between 0.8 and 1.0, then it is a mediocre model fit, and if the value for RMSEA is less than 0.8, then it shows a good fit. The resultant value for RMSEA in Table IV is 0.077, which is below the benchmark indicating a good model fit. It further proves that the value of SPI model is a better approximation of the population, and it will not be affected by the size of the sample.

5. LEAN QUALITY IMPROVEMENT MODEL

The significant quality constructs evolved through the SEM analysis and quality improvement indicators are presented in Table V. Based on these, an LQIM was evolved whose detail is given in Table VI. This conceptual detail is based on Deming's TQM philosophy of PDCA [21, 29] and SPI guidelines from literature review. The proposed LQIM is a tailored and economized model according to the practices and perceptions of the local IT practitioners. A deployment plan mapped with PDCA is also proposed and is given in Table VII.

The proposed LQIM is an indigenous lightweight quality improvement model consisting of four constructs and 10 practices that when improvised in accordance with the SMSHs cultural recommendations can establish a fit model for SMSHs. Its basic philosophy does not defy previous practices and standards but uses the predefined structures and baseline to reestablish an improvement model entirely focused on achieving a quality software product. Where a larger organization may use expensive tools and heavy investments to achieve its software product requirements and specifications, LQIM focuses more on to economizing its implementation and to ensure effective and practical results: (i) introduction of preventive measures throughout the SDLC for prevention of defect creation and its propagation; (ii) identification of new technologies (tools,

Table V. Evolved SPI paradigm practices.

LQIM constructs	Performance indicator	Variable	Reference clause (ASQ 2000-ISO 9000/CMM)
Organization size and structure	Technology, research, and development	TND	6.3/CMM-KPA-L5
Organization culture	Communication	COM	5.5/7.5.5/CMM-KPA-L3
	Training	TRA	6.2.27.5.5/CMM-KPA-L3
	Team structure	TST	4/CMM-KPA-L2
	Turnover rate	TR	5.1/CMM-KPA-L5
Organizational behavior towards quality	Software quality improvement	SQI	7.3.7/CMM-KPA-L5
	Quality assurance	QA	7.3.4/CMM-KPA-L2
Measurement and analysis	Top management quality approach	TPQ	6.1/CMM-KPA-L3
	Knowledge base library	KBL	8.4/CMM-KPA-L2,L3
	Defect log	DL	8.4/CMM-KPA-L5

L, CMM level; KPA, key process area; CMM, capability maturity model; ASQ, American Society for Quality; SPI, software process improvement; LQIM, lean quality improvement model; ISO, International Organization for Standardization.

Table VI. Lean quality improvement model conceptual detail.

Model	Variable	Indicator	Guideline description (CMM reference)
Plan	Organization size and structure	Technology research, and development	Identify new technologies (tools, methods, and processes) and track them into the organization in an orderly manner. (CMM-KPA-L5-technology change management)
	Organization Culture	Communication	Top management is required to develop effective and to easily avail communication systems that can be used by all personnel involved so that they are able to express their opinion and also express their work progress (self-monitoring) and problems/issues/violations (monitoring of others) through this system. Mode of communication can be email, phone, hand-written, and so on, but all communications are to be logged and recorded to ensure investigation of any problems and to rectify/correct any changes that do not result in a positive response. (CMM-KPA-L3-inter group coordination)
		Training	Training activities are planned, and a system for training need assessment and skill development of personnel is established to increase the adaption rate of cultural changes and personnel capabilities. (CMM-KPA-L3-training program)
		Team structure	Cross-sectional teams based on domain experts from all departments involved in LQIM development and implementation need to be established in order to ensure that all departments are part of the implementation process. (CMM-KPA-L2-software project planning)
	Organizations behavior towards quality	Top management quality approach	The top management is to provide the long-term commitment and resources to coordinate the development and maintenance of the software process quality across current and future software projects. (CMM-KPA-L3-organization process focus). It also needs to introduce preventive measures throughout the SDLC for prevention of defect creation and its propagation.
		Turnover rate	Organization needs to motivate its personnel through incentives (bonuses and titles) and trainings that can support them not only in operations but also in their career development. (CMM-KPA-L5-process change management)
Do	Organizations behavior towards quality	Quality assurance	Plan must include methods for QA so that products and services are under constant check while they are being processed therefore leaving little chances for non-conformities. (CMM-KPA-L2-software quality assurance)
		Training	Perform the trainings based on the requirements of LQIM and focused on personnel preferences and on the basis of training need assessment. Develop a learning culture through training and retraining. (CMM-KPA-L3-training program)
Check	Measurement and analysis	Quality assurance	A quality assurance system is required to ensure that the products and services meet customer requirements, and quality goals are met. (CMM-KPA-L2-software quality assurance)
		Knowledge base library	Organization is required to develop and maintain the organization's standard software process, along with related process assets, such as descriptions of software life cycles, process tailoring guidelines and criteria, the organization's software process database, and a library of software process-related

(Continues)

Table VI. (Continued)

Model	Variable	Indicator	Guideline description (CMM reference)
Act	Organizations behavior towards quality	Defect log	documentation such as process manuals, software, and operation logs for reference. (CMM-KPA-L3-organization process definition, CMM-KPA-L2-software configuration management) Defects are recorded separately so that they can be statistically analyzed to identify any trends or determine alternates to avoid/eradicate the issues. (CMM-KPA-L5-defect prevention)
		Software quality improvement	The findings based on QA activities, logs, reviews, internal audits, and statistical analysis (using various tools) need to be corrected with most appropriate corrective measure. (CMM-KPA-L5-process change management)

L, CMM level; KPA, key process area; CMM, capability maturity model; LQIM, lean quality improvement model; SDLC, software development life cycle.

Table VII. Lean quality improvement model (LQIM) deployment plan mapped with plan, do, check, act.

PLAN		PHASE I: REVIEW AND GAP ANALYSIS (Performed Once Only)	ACT →	
		Company wide LQIM review and gap analysis		
		Review of existing processes, policies and procedures		
		Identification of gaps based on best management practices		
		Submission of gap analysis report with recommendations and solution		
		TRAINING A: Introduction and Awareness on LQIM		
PLAN		DELIVERABLE: LQIM Gap Analysis and Recommendation Report		←
		PHASE II: SYSTEM DEVELOPMENT / REVISION / IMPROVEMENT		
	↓	Development of quality policy and quality objectives		
		Development of process flow charts, corporate organization chart and departmental charts		
		LQIM development and implementation monitoring team		
		Development of technology development and research & development department		
		Development of procedure for corrective and preventive measures		
		Development of HR policies (to reduce turnover rate)		
		Development of knowledge base library		
		Development of procedure for control of non-conformance		
		Development of procedure fro quality assurance		
		Development of procedures for communication		
DO		Development of training and awareness procedures and plans (systematic culture change acceptance)		
		Development of procedure fro internal system auditing (development of audit checklist)		
		TRAINING B: Documentation, Recording and Reporting based on LQIM Requirements		
		DELIVERABLE: LQIM Policies, Procedures, and Templates		
DO		PHASE III: IMPLEMENTATION		
	↓	Implementation team assignment		
		Implementation of records as per developed procedures		
		Implementation of records and awareness verification relating to LQIM		
		Correction of documentation based on feedback		
CHECK		DELIVERABLE: LQIM Manual	→	
		PHASE IV: AUDITING		
	↓	Selection of internal auditors		
		Planning the internal audit		
		Collection of defect logs and updating knowledge base library		
		Company-wide LQIM internal audit		
		Corrective and preventive actions / audit non-conformity closing		
		Trend analysis on the basis of previous audit results		
CHECK		TRAINING C: Internal Audit Training		
		DELIVERABLE: Internal Audit Report		

methods, and processes) and tracking them into the organization in an orderly manner; (iii) development of effective and easily available communication systems that can be used by all personnel involved; (iv) development of an organizational level learning culture; (v) establishment of a system for training need assessment and skill development; (vi) commitment of the organization to give incentives to their employees and to support them in their career development; (vii)

establishment of cross-sectional teams based on domain experts from different departments; (viii) long-term commitment of the organization towards development and maintenance of software process quality; (ix) planning for quality assurance; (x) establishment of a quality assurance system; (xi) defining, developing, and maintaining organizational level standard software process assets; (xii) establishment of a library of organizational process assets; (xiii) establishment of a system of recording and analyzing defect data; and (xiv) taking corrective measures based on the analysis of the defect data collected.

Compared with other quality models such as OWPL, software process matrix, PRISMS, and improvement framework utilizing lightweight assessment and improvement planning for small organizations, which are based on process-oriented approach, the LQIM is based on TQM philosophy and quality culture. LQIM proposes preventive measures early in the SDLC to prevent defect creation and propagation as compared with the corrective approach for defect management and prioritization as followed by other models. The LQIM follows lightweight economical Deming's PDCA cycle for continuous process improvement, whereas others require expensive assessment techniques like software process improvement and capability determination and experienced assessors. LQIM emphasizes more on cross-functional teams for PQI from all the departments within the organization compared with other similar models. LQIM proposes to establish knowledge base library for data collection and update, whereas other models do not mention any automatic data collection process [15]. Many of the quality models for small organizations require experienced assessors, whereas LQIM implements process improvement through cross-functional teams. LQIM does training need assessment based on process and culture changes, whereas other models focus mainly on competency-based training. An important feature of LQIM is to establish freedom of speech as part of cultural change through the establishment of open communication channels.

Lean quality improvement model has its limitations too. The LQIM paradigm shift involves culture and process changes. Such process reengineering efforts result in resistance to change and hence are prone to low level of acceptance. Hesitation in employee empowerment to delegate decisions may also hinder the implementation of LQIM. Similarly, the process slack time may increase as not all personnel can be part of skill development and training programs because of limitation of resources. Also management may lack consistency in system improvement and planning in the long run, and during such intervals, customers may prefer (and enforce in some cases) their own process improvement policies over LQIM policies. Resource allocation is another problem as departments obtain resources according to their priority; therefore, some departments are ignored intentionally in the beginning. Absence of TQM mindset and quality culture may lead to schedule delays, cost overruns, and higher rework costs.

6. LEAN QUALITY IMPROVEMENT MODEL VALIDATION THROUGH EXPERT OPINION

To validate the LQIM model, developed expert opinions were obtained. A questionnaire was developed with reference to the study conducted by Beecham *et al.* [105]. The survey was personally administered among industrial experts who were experienced and involved in software quality assurance and improvement. Senior faculty members teaching software engineering and area experts in TQM were also part of the survey. The panel included 14 practitioners, five academics, and one of those was involved in both activities. The average experience of the experts interviewed was 10 years. The expert was first briefed by the researcher about the LQIM model, its adaptation from ISO 9000 and CMM and terminologies including abbreviations.

Each of the constructs was measured on the 5-point Likert scale from 1=strongly disagree to 5=strongly agree. The mean values of the responses for adherence to quality improvement characteristics, scope, understandability, ease of use, and tailorability constructs are 3.8875, 3.6250, 3.7300, 3.5500, and 3.8750, respectively, with the respective standard deviation of 0.48310, 0.82518, 0.54782, 0.74162, and 0.72321. This indicates an overall 'agreement' with the proposed LQIM model. Table VIII shows that all of the constructs have the value of Cronbach's alpha above 0.6, which is accepted according to Nunnally [93]. The overall Cronbach's alpha for this study is

Table VIII. Validation survey results - $N=20$.

Constructs	Mean	Standard deviation	Cronbach's alpha
Adherence to quality improvement characteristics	3.8875	0.48310	0.629
Scope	3.7300	0.82518	0.680
Understandability	3.5500	0.54782	0.720
Ease of use	4.1000	0.74162	0.612
Tailorability	3.8750	0.72321	0.634

0.841, which is acceptable under the findings of Nunnally and Bernstein [106] and Gefen *et al.* [107]. All of the items are retained except for the ease of use. There were three items for this construct, but only two items were showing significant correlation and are retained to obtain Cronbach's alpha of 0.612, which is within the acceptable limits.

During the interviews, the following comments were also received: (i) the 'turnover rate' at the plan level shall be taken as 'target turnover rate' and shall be measured at the check level; (ii) in the plan level, emphasis shall be made to complete the project on-time, within schedule; and (iii) plan level shall be divided into organizational level planning and project planning. In short, this study has identified that (i) the LQIM confirms adherence to quality characteristics; (ii) its scope covers minimal industrial standards for quality improvement; (iii) it can be easily tailored; and (iv) it is easy to use. Hence, the expert opinions thus received through this validation survey support the LQIM.

7. CONCLUSIONS AND FUTURE WORK

This paper presents an LQIM and a deployment plan to implement total quality culture and standard quality practices in the software industry in Pakistan. The proposed LQIM paradigm is a lightweight baseline model for an SME that is starting to implement quality practices. As shown in Table V, the LQIM is based on four constructs and 10 practices whose conceptual detail and deployment plan are given in Tables VI and VII, respectively. The LQIM is validated using good fit indices in SEM and also through expert opinion survey. Implementation of indigenous LQIM is recommended using the Deming's TQM philosophy of PDCA cycle for continuous process improvement. The indigenized LQIM is destined to give innovative and flexible directions for SSMH in Pakistan to change their culture from corrective to preventive quality culture and to improve their processes economically by following TQM philosophy.

In future, the study can be extended by considering different types of variables such as development domain and operating model. Also the research can further be replicated in other developing countries to develop the LQIM that caters the needs and cultural requirements of SMEs in the respective developing countries.

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